SKILL-1

Name: B. Vida Swetha
Roll Number: 2520030100

Observation Table

Parameter	TCP Packet	UDP Packet
Frame Length	54 bytes	132 bytes
Ethernet Header Length	14 bytes	14 bytes + 4-byte 802.1Q VLAN tag
IP Header Length	20 bytes (IPv4)	40 bytes (IPv6)
Transport Header Length	20 bytes (TCP)	8 bytes (UDP)
IP Total Length	40 bytes	114 bytes (IPv6 header + IPv6 payload)
Payload Length	0 bytes	66 bytes (calculated from captured frame)

Header Overhead Calculation

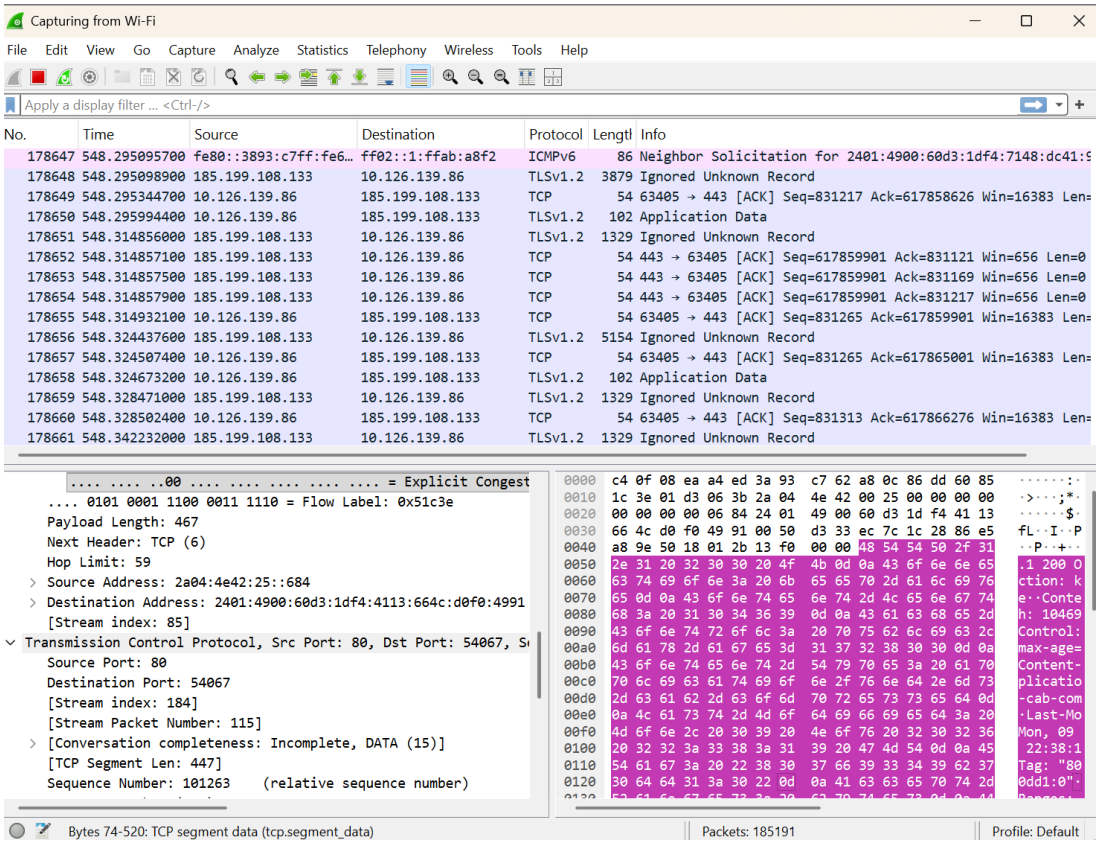
TCP Packet:

- Frame Length = 54 bytes
- Header Overhead = 14 + 20 + 20 = 54 bytes
- Payload = 0 bytes
- Payload Efficiency = (0/54) × 100 = 0%

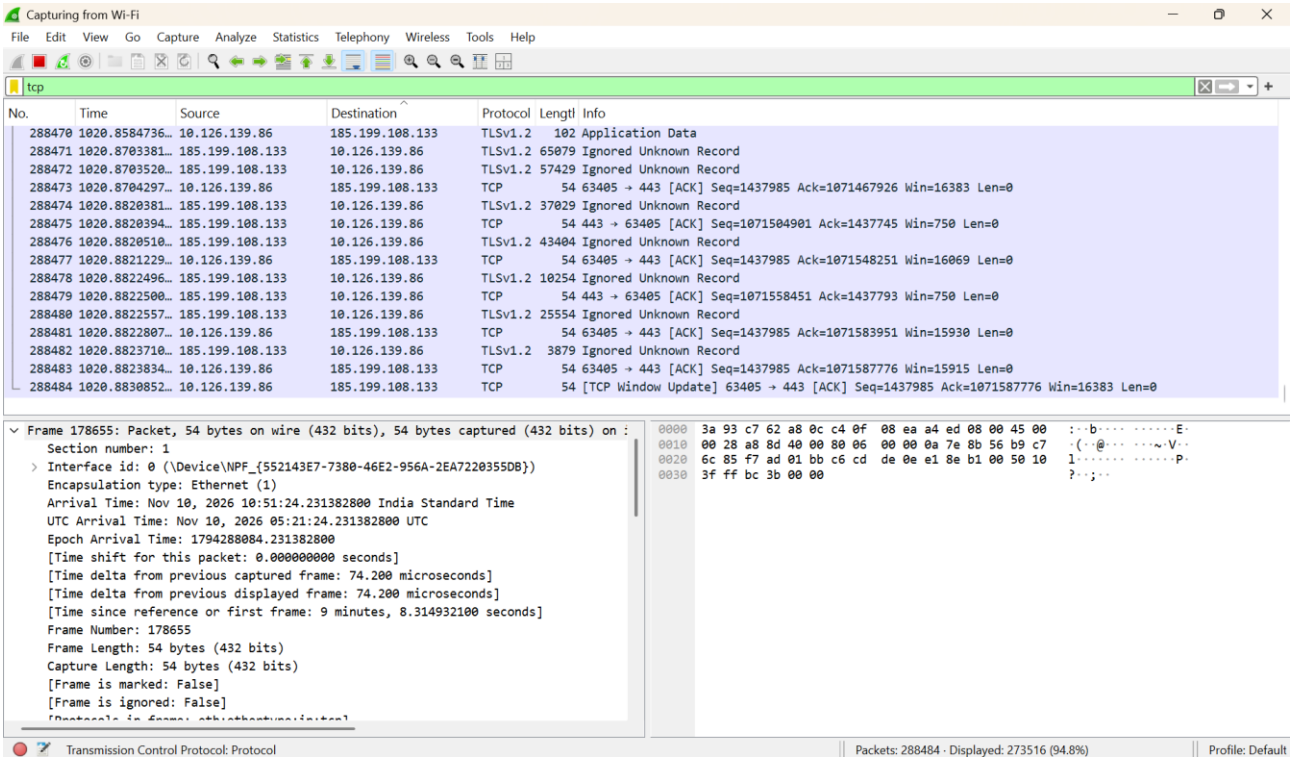
UDP Packet:

- Frame Length = 132 bytes
- Header Overhead = 18 + 40 + 8 = 66 bytes
- Payload = 66 bytes
- Payload Efficiency = (66/132) × 100 = 50%

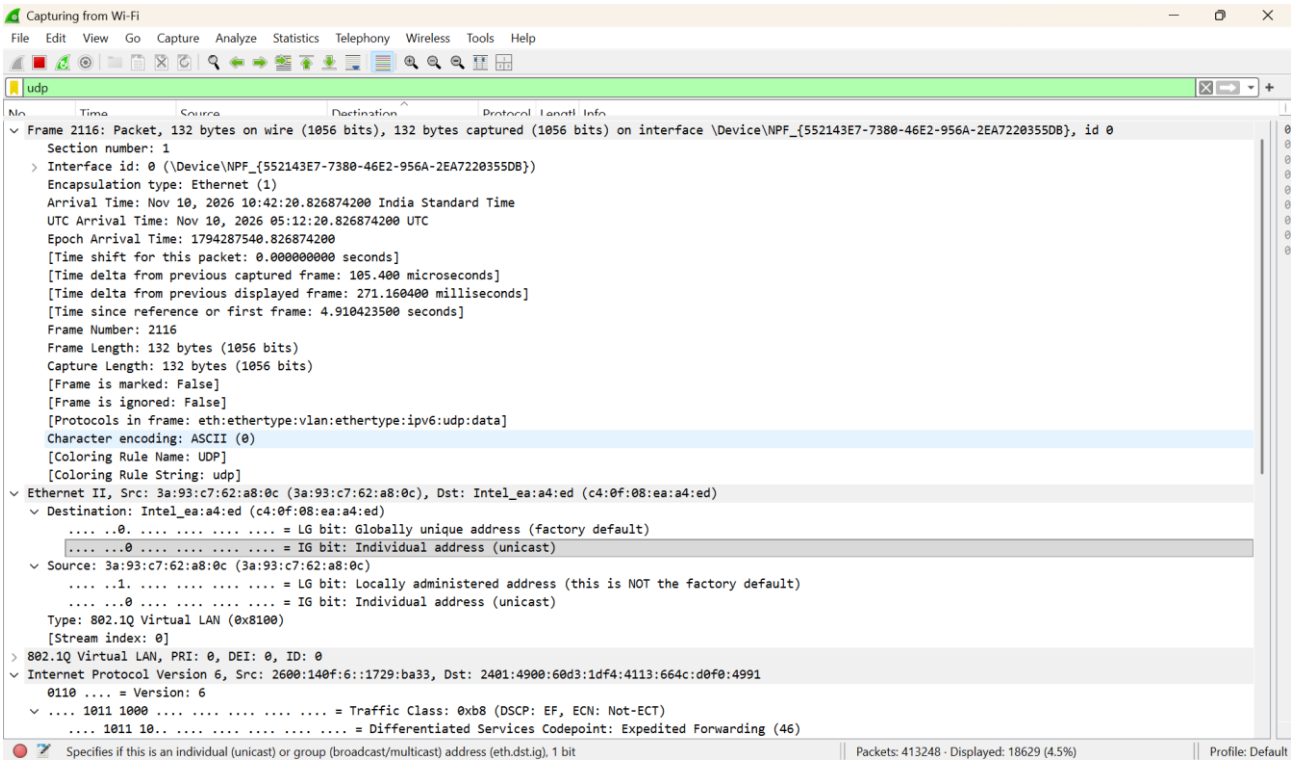
- Wireshark Screenshots



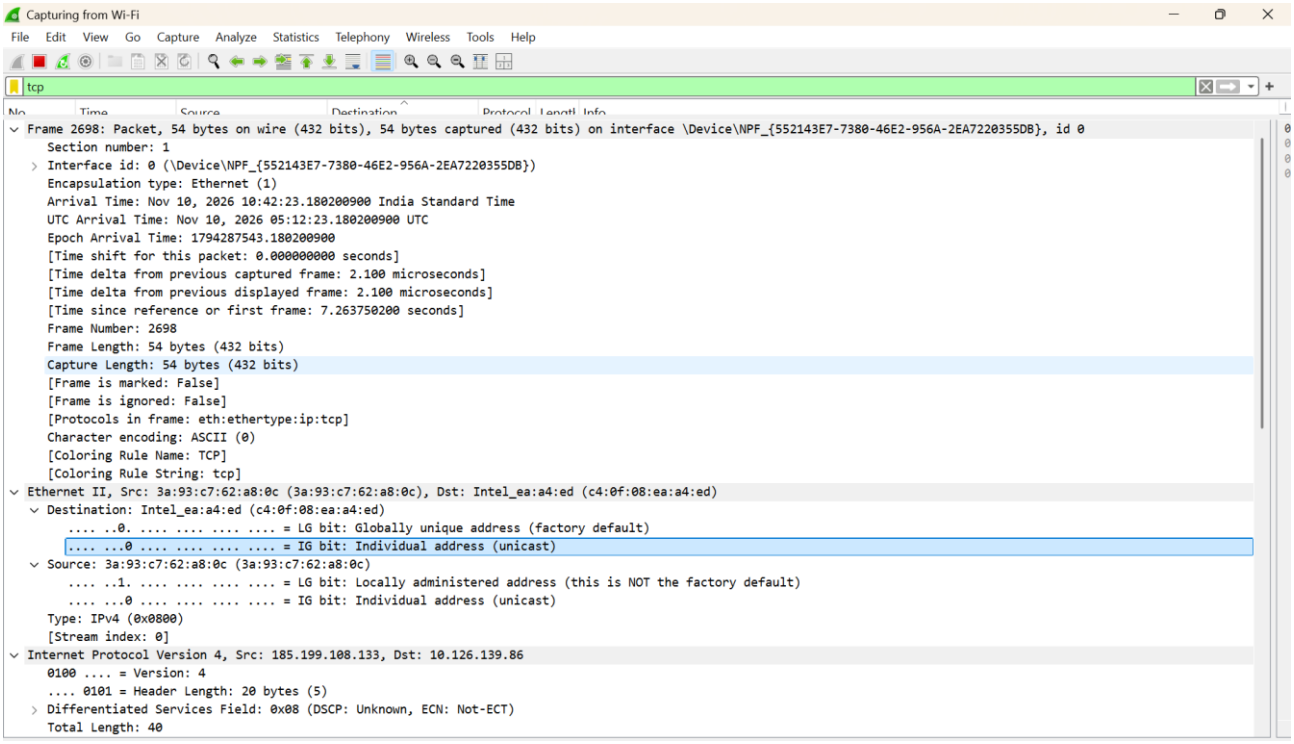
Wireshark packet capture showing captured traffic and packet details.



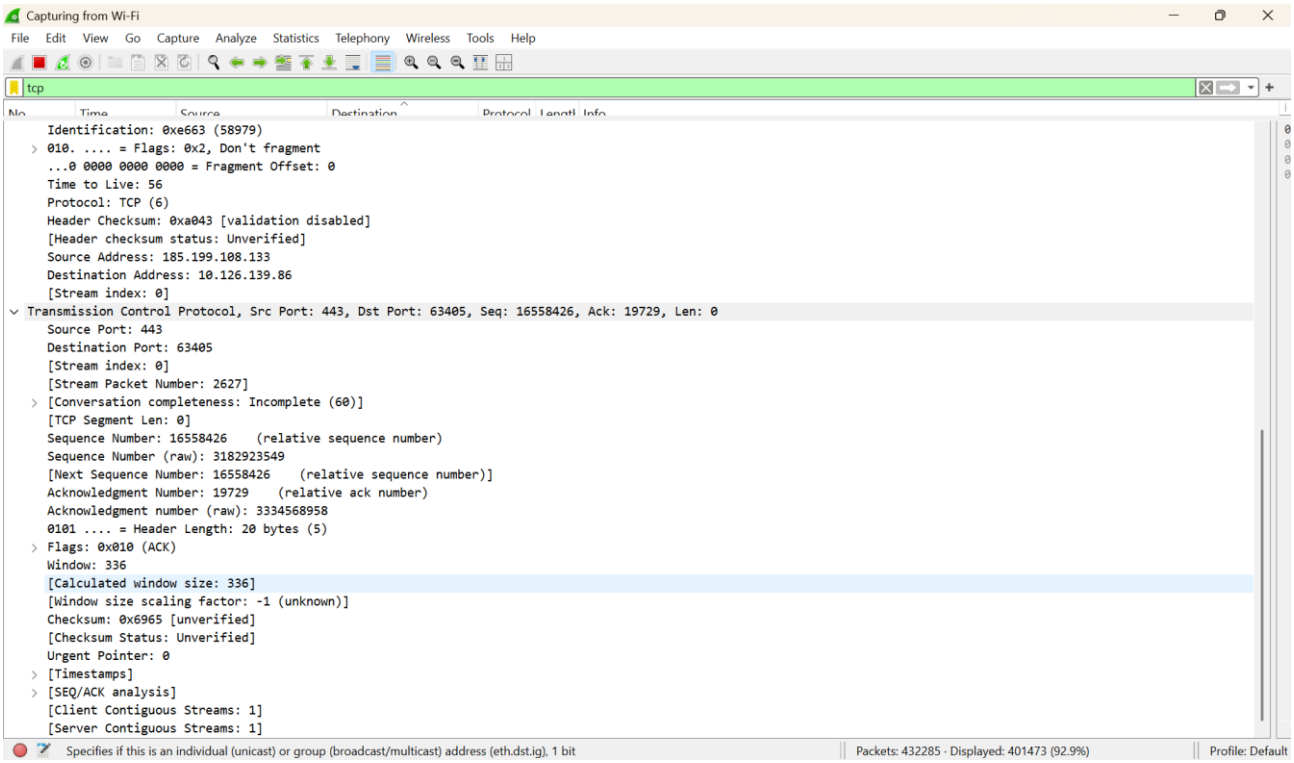
TCP display filter applied; packet list shows TCP/TLS traffic.



UDP packet details showing Ethernet, VLAN, IPv6, and UDP protocol layers.



TCP packet details showing Ethernet II and IPv4 information.



TCP header details, including source/destination ports, ACK flag, and zero TCP payload.